

Assignment No. 1

ScreenShots

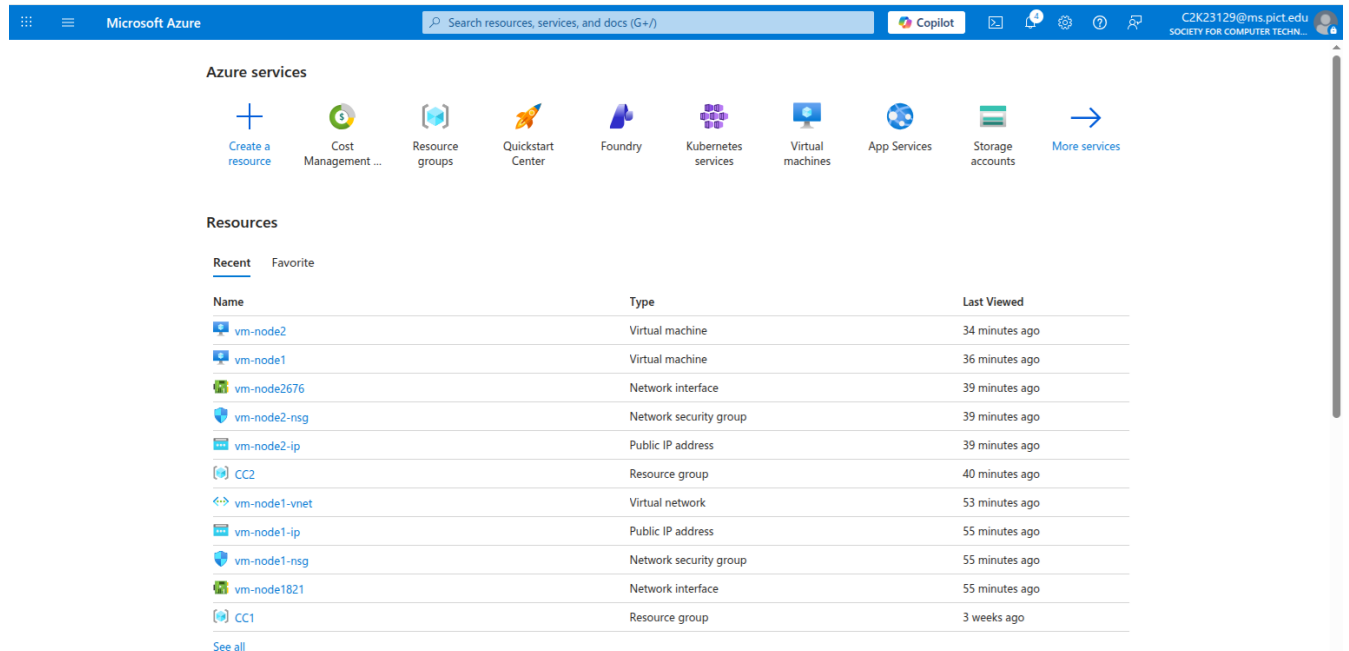


Fig 1. Overview of Azure Resources Including Virtual Machines, Network Interfaces, and Security Groups in the Microsoft Azure Portal used in assignment

Procedure

5. Procedure (Step-by-Step)

Describe the workflow in a professional, narrative style

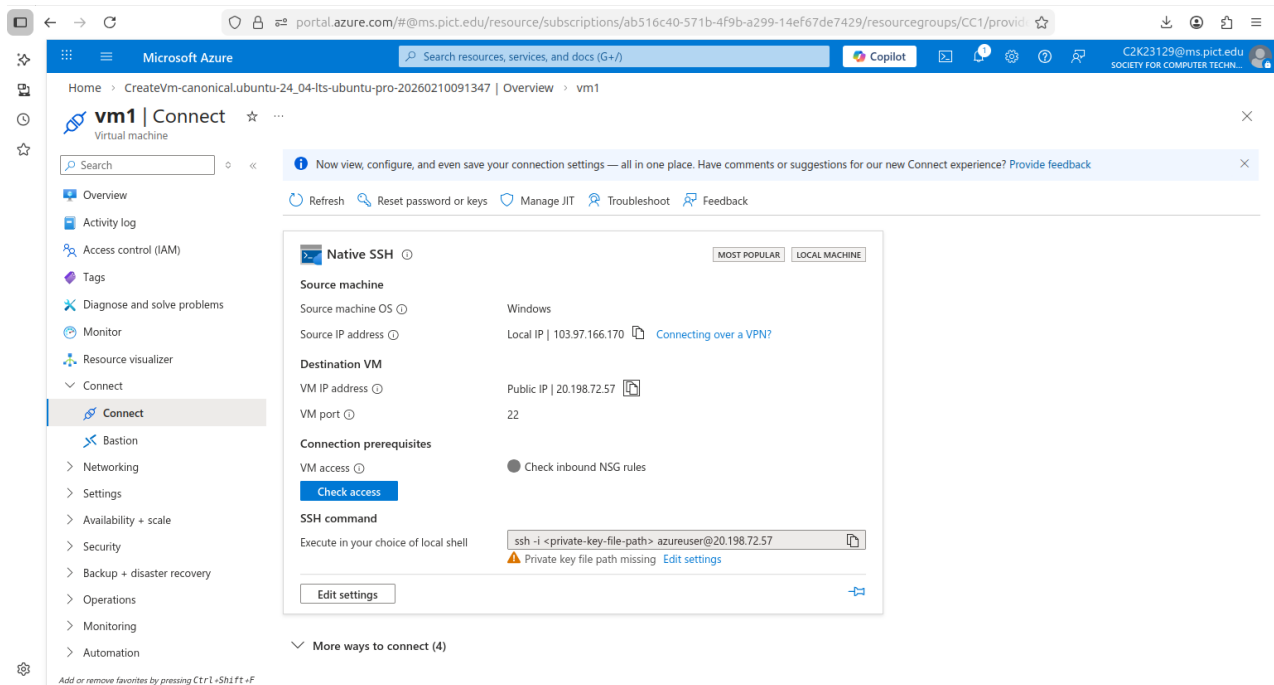
- **Phase 1: Environment Setup** (Creating the instance/VPC).
 1. Log in to the **Microsoft Azure portal**.
 2. Create a **Resource Group** to manage the resources.
 3. Create a **Virtual Machine (VM)** that will act as the server.
 4. Select the operating system (such as Linux).
 5. Configure the **Public IP address** and allow **HTTP (port 80)** in network settings.
- **Phase 2: Configuration** (Installing software, setting permissions).
 1. Connect to the virtual machine using **SSH**.
 2. Update the system packages.
 3. Install a web server such as **Apache HTTP Server**.

4. Start and enable the web server service.
5. Verify that the server is running by accessing the VM's public IP address.

- **Phase 3: Deployment** (Uploading code, starting services).

1. Upload the HTML file to the server using **SCP**.
2. Place the file in the web server directory (for example `/var/www/html`).
3. Ensure the main file is named **index.html**.
4. Restart the web server if required.
5. Open the public IP address in a browser to view the hosted HTML webpage.

Observations and Results



- Fig 1. Virtual Machine Created in Microsoft Azure Portal

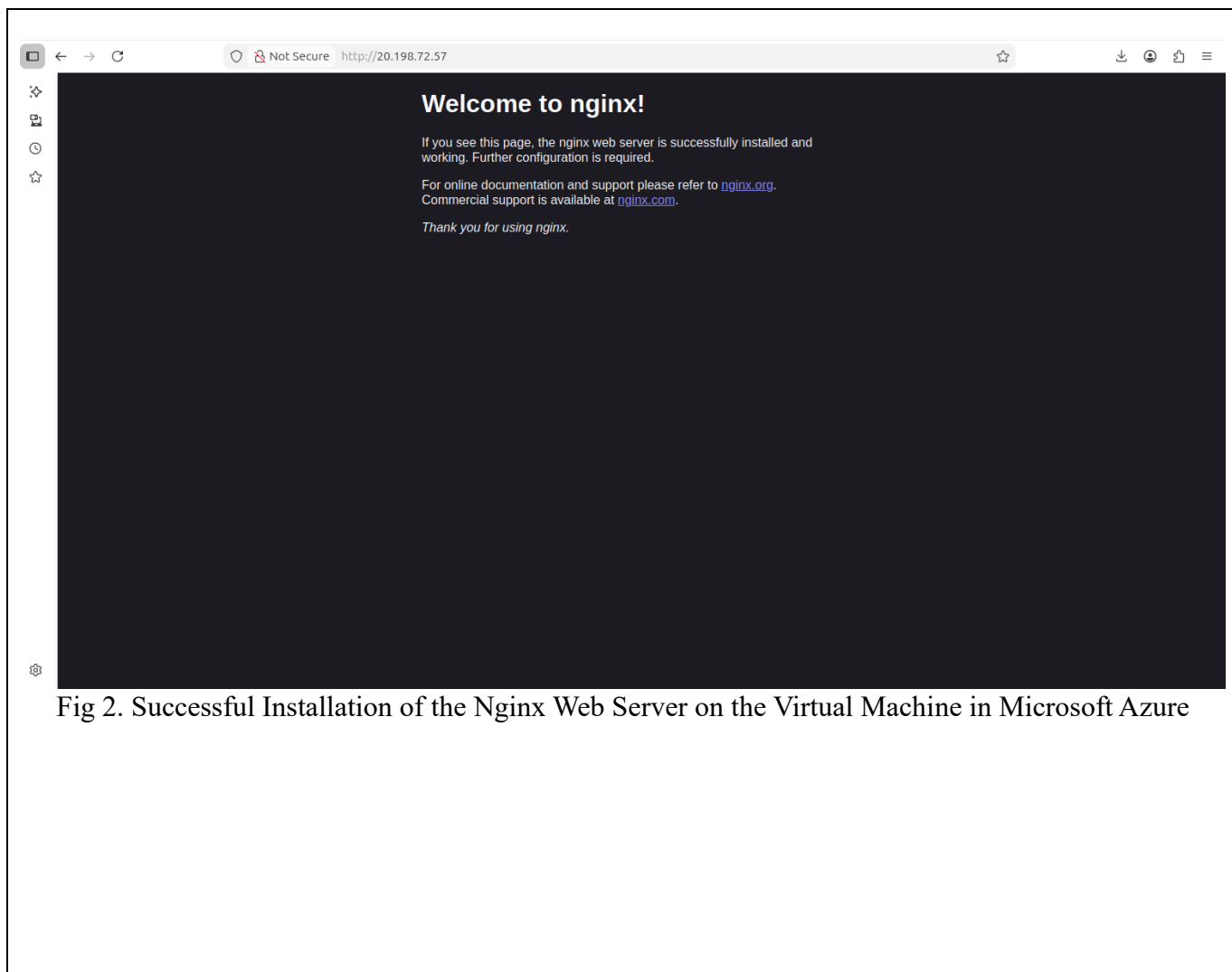


Fig 2. Successful Installation of the Nginx Web Server on the Virtual Machine in Microsoft Azure

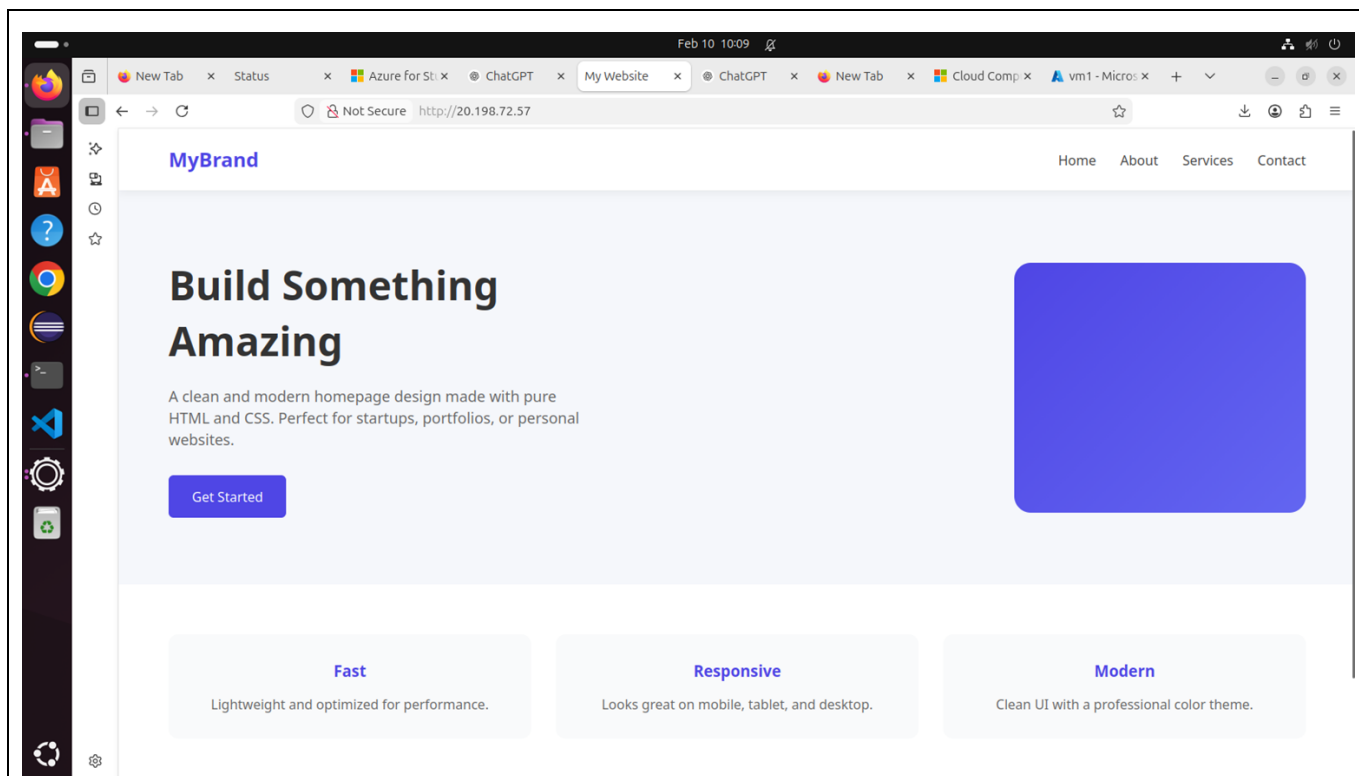


Fig 3. Successfully Hosted HTML Website on a Virtual Machine using Nginx Web Server in Microsoft Azure

Analysis

- The HTML website was successfully deployed and accessed through the public IP address of the virtual machine hosted on Microsoft Azure.
- The Nginx web server correctly served the HTML content, confirming that the server configuration and file placement were properly completed.
- Hosting the website on Azure allowed the webpage to be accessible over the internet, demonstrating how cloud platforms can efficiently host and manage web applications.

Conclusion

In this assignment, a basic HTML website was successfully hosted using the cloud services provided by Microsoft Azure. A virtual machine was created and configured to act as a web server environment. The Nginx web server was installed and properly configured to serve web content. After uploading the HTML file to the server directory, the website was successfully deployed and accessed through the public IP address of the virtual machine.